CAD ENGINEERING TECHNICAL INTERVIEW

EDA / CAD Engineering Projects

Logic Synthesis / Physical Design / Timing ECO / CAD Flow Automation

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Education & Research Interests

Education

National Taiwan University
M.S. Candidate, Data Science Master Degree Program
Sep. 2025 – Expected Jun. 2027

National Chengchi University
B.S. in Computer Science
Double Major in Electronic Physics
Sep. 2021 – Jun. 2025

Research Interests

EDA / CAD & VLSI Design Automation

Logic Synthesis
Technology Mapping
Static Timing Analysis
Timing Closure
Physical Design
Incremental ECO
PPA

Algorithm & Software Development

C++ / Python / Tcl Flow Automation

Professional Software Engineering Experience

Details index

Software development, simulation, and validation experience in R&D environments.

AIMobile Co., Ltd. (Inventec Group)

Apr. 2025 – Jul. 2025

Contract Software Engineer

Contributed to software development and system integration in an R&D environment.

Supported debugging and functional validation across software modules.

Neopower Technologies Co., Ltd.

Jun. 2023 – Mar. 2025

Contract Software Engineer

Developed a custom system simulator to reproduce operational and error scenarios.

Performed regression checks, log analysis, and customer-side A/B validation.

Selected Projects & Discussion Topics

Details index

03 / PROJECT OVERVIEW

01 Master's Research — Logic Synthesis (Front End)

Joint Single- and Multi-Output Cell Mapping via Differentiable Optimization

EDA Tool Development / Boolean Matching / Area–Delay Optimization

02 ICCAD 2025 Problem C — CAD Methodology (Back End)

STA-Driven Incremental Timing ECO Search

OpenROAD / OpenSTA / Python–Tcl Flow Automation / Heuristic PPA Search

PROJECT 01 / MASTER'S RESEARCH / LOGIC SYNTHESIS

Technology Mapping for Multi-Output Standard Cells

FA/HA cells over a shared input boundary

MY PROPOSED APPROACH

Joint Single- and Multi-Output Cell Mapping via Differentiable Optimization

NF-direct MO discovery in ABC, area-aware MILP pre-selection, and GradMap.

Joint SO / MO Mapping: Research Pipeline

01 / RESEARCH SCOPE

Details index | SO baseline | GradMap integration

Discover FA/HA opportunities, then pre-select non-overlapping MO candidates before GradMap.

Problem

Single-output cut retention does not target two-output FA/HA opportunities.

NF-Direct Discovery

Independent MO cut search inside ABC `&nf`. SO mapping stays unchanged.

MILP Pre-selection

Keep a non-overlapping, area-aware subset of legal MO candidates.

FIGURE SPACE

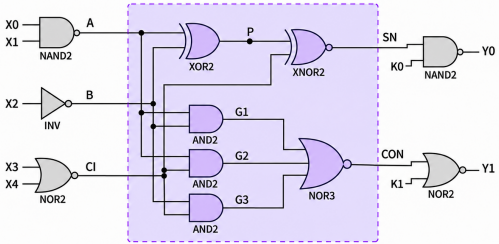
Joint SO / MO mapping framework

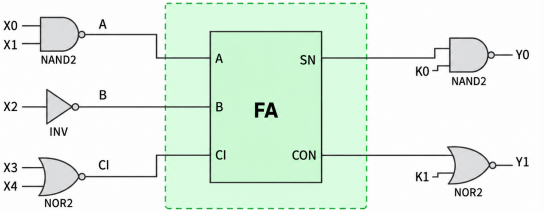
SO matches stay available. Legal MO candidates go through MILP, then GradMap chooses SO or BIND.

Single-Output vs Multi-Output Cell Mapping

03 / MOTIVATION

A shared FA cell can implement two output functions over the same input boundary.





Independent MO discovery exposes joint output matches beyond SO-oriented cut retention.

NF-Direct MO Candidate Discovery

Details index | Pair legality | SO witness

Pair compatible CON and SN projections over a common input support.

01 / MO Cut Propagation

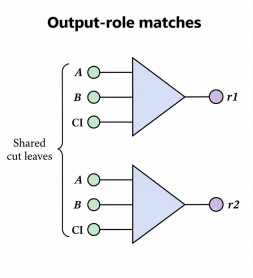
Independent retention of FA/HA-relevant truth functions and supports.

02 / Projection Pairing

Match CON / SN roles using shared leaves, permutation and phase.

03 / Legality + Witness

Validate the pair and export v3 MOCAND records with replacement roots.



r_1, r_2: complementary output-role matches; common leaves A, B, CI.

OUTPUT: BIND / ROOTS / FANINS / REPLACE_ROOTS

Baseline: File-Order Candidate Loss

Details index GradMap integration

File-order greedy filtering ignores estimated area benefits of conflicting MO candidates.

41,476
Original MO candidate BIND records

7,024
FIFO retained (offline simulation)

16.9%
Overall retention rate

Design	Original MO	FIFO Retained	Dropped	Retention
EPFL hyp	25,440	4,180	21,260	16.4%
EPFL square	3,658	544	3,114	14.9%
EPFL log2	2,798	400	2,398	14.3%
EPFL multiplier	2,190	349	1,841	15.9%
EPFL voter	632	106	526	16.8%
ISCAS c6288	258	37	221	14.3%
ISCAS c7552	102	11	91	10.8%
EPFL adder	6	1	5	16.7%
All 364 designs	**41,476**	**7,024**	**34,452**	**16.9%**

Source: NF-direct full-corpus census, 364 benchmark entries; simulated parser-policy first-fit, not trained GradMap QoR.

Area-Aware MO Selection (MILP)

Maximize estimated area savings subject to non-overlapping MO candidates.

$$\begin{aligned} \max \quad & \sum_{i=1}^{N} \left(A_i^{SO} - A_i^{MO}\right) x_i \\ \text{s.t.} \quad & \sum_{i=1}^{N} a_{vi} x_i \le 1, \quad \forall v \in \mathcal{V} \\ & x_i \in \{0,1\}, \quad i = 1, \ldots, N \end{aligned}$$

FIGURE SPACE

Conflict: (A,B) vs (A,C)

Shared AIG node A, so MILP keeps at most one candidate.

Implementation

Python + HiGHS (`highspy`); emit selected.matches.

Notation

x_i: select MO i; A_i^{SO}: SO witness area; A_i^{MO}: MO cell area;

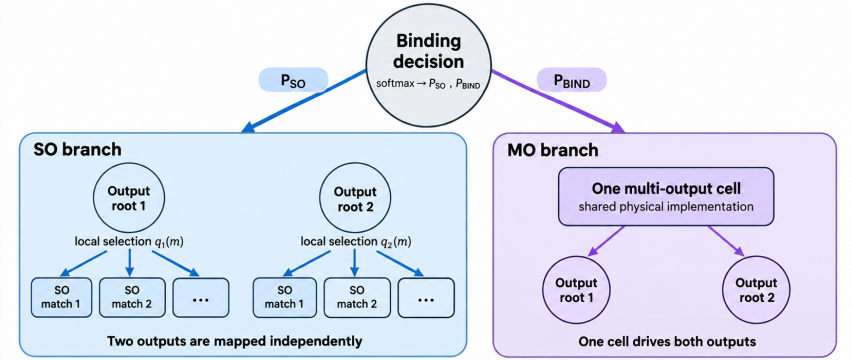
$a_{vi} = 1$ iff AIG base node v belongs to MO candidate i; otherwise 0.

Objective weights estimate local SO-to-MO benefits; final QoR is evaluated after GradMap mapping.

Connecting `&nf -Y` to GradMap

Details index Verification

MILP passes non-overlapping MO candidates to GradMap, which still chooses SO or BIND.



MO area is counted once. Timing is read at each output pin.

Current Mapping Flow & ADP Improvement

11 / CURRENT RESULTS

Current flow

NF-direct MO discovery

Boolean / structural legality

SO witnesses + area weights

MILP candidate pre-selection

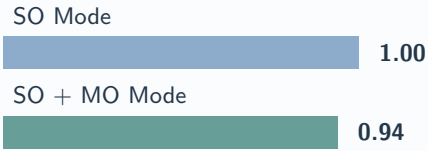
GradMap SO / MO optimization

Measure final mapped QoR after optimization; MILP weights are local estimates.

6%

Lower ADP vs. SO Mode

Normalized ADP (SO Mode = 1.00)



ADP = area × delay.

PROJECT 02 / ICCAD 2025 CAD CONTEST / PROBLEM C

Incremental Placement Optimization Beyond Detailed Placement

Simultaneous Gate Sizing, Buffering, and Cell Relocation

MY PROPOSED APPROACH

STA-Driven Incremental Timing ECO Search

Use OpenROAD / OpenSTA with policy-controlled ECO moves and global PPA evaluation.

Multi-Objective Timing ECO Optimization

02 / PROBLEM DEFINITION

Details index | STA targets | QoR / budget

Improve setup TNS toward zero while controlling total power, HPWL, and cell displacement.

OPTIMIZATION OBJECTIVES

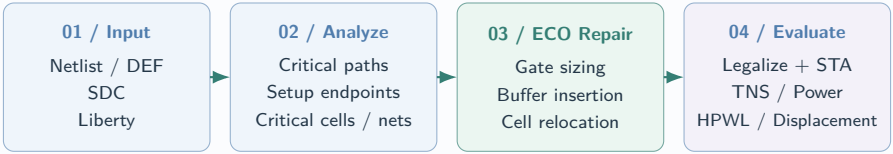
01 / Setup TNS
Reduce negative setup slack at violating endpoints.

02 / Total Power
Avoid excess power from ECO changes.

03 / HPWL
Control estimated wirelength after ECO.

04 / Displacement
Limit cell movement and keep placement legal.

BASIC ECO EVALUATION PIPELINE



Key challenge: a timing fix can worsen power or physical cost.

Five Automatic ECO Strategy Candidates

02 / CANDIDATE SET

Details index | STA targets | Endpoint coverage | ECO sequences

The candidate is an automatic repair policy and editable region, not a manually chosen cell replacement.

S1 / Global Setup

Whole design; automatic upsizing and VT selection.

S2 / Path-Restricted

Top-K path cells + nets; localized setup repair.

S3 / Overlap-Aware

Top-I shared cells, nets fixed; sizing / VT repair.

S4 / Local Buffering

Top-K nets, original cells fixed; buffer insertion and load splitting.

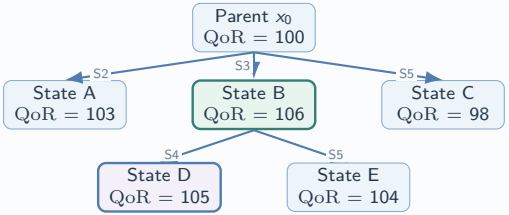
S5 / Power Recovery

Slack-rich cells / wider region; lower-power equivalent cells.

STA-Guided ECO Search: Best-First Branch List

No better child this round? Continue from the best unexplored state in the Branch List.

Search Tree Example



Bounded Branch List: D:105 > E:104 > A:103

Global Best: B (106) | **Next:** D (105)

A lower-scoring intermediate state may lead to a better ECO sequence later.

Heuristic Search Steps

01 / Pop Parent

Highest-QoR unexplored state from the **Branch List**.

02 / Generate + Evaluate

STA-guided ECO moves, legalization, and global QoR score.

03 / Update States

Save improved **Global Best**; queue legal child states.

04 / Next Branch

Expand best branch, **even with no new Global Best**.

QoR Results & Conclusion

02 / EARLIER ECO EVALUATION

Setup-focused timing repair and the resulting power / wirelength trade-off.

Setup Timing

16.7%
Median Setup TNS Improvement

14.6%
Average Setup WNS Improvement

Power & Wirelength

8%
Total Power Reduction

+2%
HPWL Increase

Conclusion

ECO operators can improve setup timing and power while accepting a small wirelength trade-off.

THANK YOU

Questions & Discussion

▸ Technical details

Technical Details

Back to project overview

QUESTIONS & DISCUSSION

Select a topic. Each detail page links back to the relevant summary slides.

Multi-Output Technology Mapping

D01 / SO baseline

D02 / Pair legality

D03 / SO witness

D04 / Verification

STA-Driven Timing ECO

D05 / STA targets

D06 / Endpoint coverage

D07 / ECO sequences

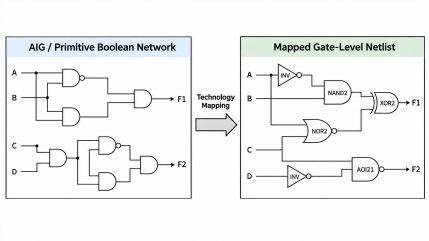
D08 / QoR / budget

D09 / Parallel workers

ABC &nf: The Single-Output Baseline

Preserve original NF SO behavior while exposing its candidate search space.

AIG + Liberty → SO cut enumeration → Delay / Area mapping → C / M match dump



Same Boolean behavior, realized with library cells.

Classic C Records

All retained single-output cut × library-cell matches.

Classic M Records

The final SO cover selected by ABC NF.

Invariant: enabling NF-direct MO discovery preserves classic C/M, vCellMapping, SO area and delay.

MO Pair Matching & Structural Legality

05 / CANDIDATE VALIDATION

Boolean matching is necessary, but not sufficient, for legal MO binding.

Boolean Matching

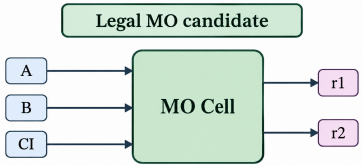
CON and SN roots match the cell's two truth-table projections.

Physical Compatibility

Same input support and a compatible pin permutation / polarity.

Structural Legality

Distinct roots; reject direct-dangling and TFI / cycle conflicts.



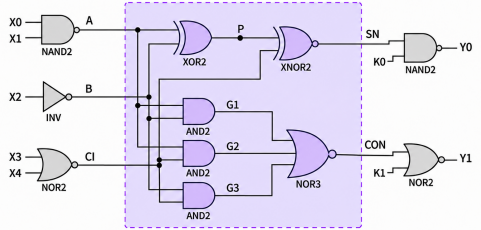
Before MILP

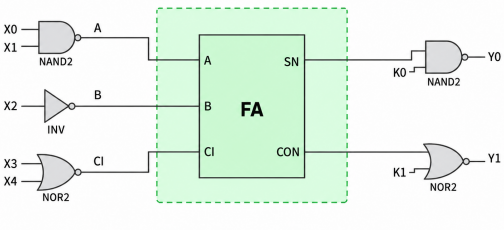
Only individually legal, witness-backed MO records enter global selection.

SO Witness: Replacing Multiple SO Cells

06 / REPLACEMENT SEMANTICS

An MO candidate replaces a multi-level SO subcover with two output roots.





Trace the SO witness to the common MO input boundary. Record participating nodes as `REPLACE_ROOTS`; preserve any internal node still demanded externally.

The SO witness need not equal ABC's final selected SO cover.

Python + HiGHS: Verification Pipeline

Separate pre-selection quality from GradMap final mapped area.



Correctness Checks

Gate A: unchanged original NF mapping
Gate B: individual MO legality
Gate C: GradMap materialization + CEC

Experimental Comparison

SO-only vs FIFO vs MILP-selected MO; same AIG, library, GradMap settings, and seed.

FIGURE SPACE

ADP and runtime chart

SO-only, FIFO, and MILP-selected MO, under the same settings.

Timing-Driven Target Extraction

Derive editable cells and nets from setup-violating paths and distinct endpoints.

```
report_checks -path_delay max
     -group_count 200
     -endpoint_count 3
     -slack_max 0
```

STA Report Scope

Setup paths may be PI→FF, FF→FF, FF→PO, or PI→PO, depending on constraints.

1 / Extract

Record endpoint, slack, path instances and nets.

2 / Deduplicate

Do not count the same violating endpoint repeatedly.

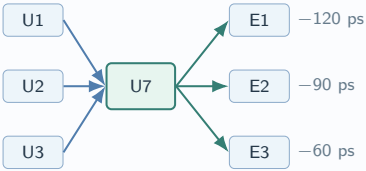
3 / Prioritize

Rank shared cells, critical nets, and candidate repair regions.

The STA report chooses where to search; OpenROAD performs the actual repair operation.

Slack-Weighted Endpoint Coverage

Prioritize shared cells by distinct endpoint coverage and violation severity.



One shared cell influences three distinct setup endpoints.

Distinct Endpoint Coverage

$Coverage(c) = |E(c)|$

Each endpoint is counted only once.

Slack-Weighted Priority

$$Priority(c) = \sum_{e \in E(c)} \max(0, -slack_e)$$

This priority guides search; the real ΔTNS still requires global STA after ECO.

Sequential ECO Strategy Combinations

Build a candidate by chaining compatible automatic repairs from a saved checkpoint.

S3 → S5

Repair shared timing-critical cells, then attempt power recovery if slack permits.

S4 → S3

Buffer selected critical nets, then re-target the remaining critical cells.

S3 → S4

Size critical cells first, then address residual net-delay bottlenecks.

Candidate Execution

Parent checkpoint

↓

Automatic repair pass 1

↓

Optional repair pass 2

↓

Legalize + re-evaluate global QoR

Operator order matters: buffering changes load; sizing changes driver strength.

Checkpoint Quality and Runtime Budget

Rank legal ECO checkpoints with consistent global QoR references, under one runtime budget.

ILLUSTRATIVE INTERNAL RANKING SCORE

$$\boxed{Q(x) = w_T\, g_T(x) + w_P\, g_P(x) - w_H\, p_H(x)}$$

$$g_T = \frac{|TNS_0| - |TNS_x|}{\max(|TNS_0|, \epsilon)}, \quad g_P = \frac{P_0 - P_x}{P_0}, \quad p_H = \frac{H_x - H_0}{H_0}$$

Quality-Based Search

Use the same baseline and weights when evaluating different branches; reject illegal or failed ECO states.

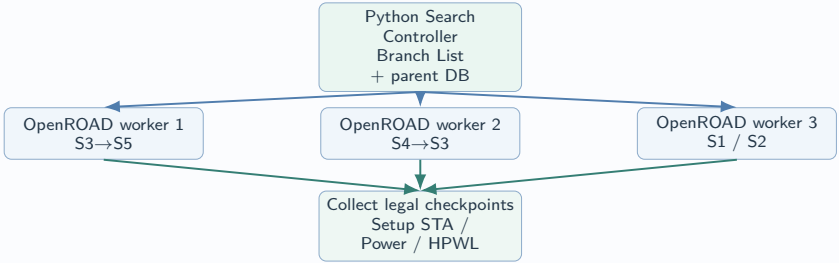
Runtime-Aware Execution

Batch processing consumes wall-clock time once; stop when the global execution budget is exhausted.

TNS: setup total negative slack; *P*: total power; *H*: HPWL. Weights define a search heuristic, not an official contest score.

Parallel Candidate Evaluation with OpenROAD

Independent workers run candidate ECO sequences from a selected Branch List checkpoint.



Worker Responsibilities

Apply ECO policy, legalize placement, update parasitics, collect full-design QoR, and save a restorable checkpoint.

Controller Responsibilities

Bound workers by available resources; update global best and Branch List using consistent QoR results.